

JOSEPH DAYNGER RUIZ

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CURRICULUM VITAE

EDUCATION

Doctorate Degree in Progress

Fall 2021 – Present

University of Arizona, Tucson | GPA: 3.871 / 4.0

Courses included: Abstract Algebra, Algebraic Number Theory, Algebraic Geometry, Real Analysis, Topology-Geometry, Commutative Algebra, Global Differential Geometry, Information Theory/Machine Learning, Group Theory, Complex Analysis, Geometric Invariant Theory, Homological Algebra, Computational Linguistics, Theory of Linear Systems.

Bachelor of Science in Mathematics

Fall 2017 – Spring 2020

Metropolitan State University of Denver, Denver | GPA: 3.49 / 4.0

Courses included: Abstract Algebra, Topology, Algebraic Topology, Complex Variables, Chaos Theory, Differential Equations, Partial Differential Equations, Probability and Statistics, Representation Theory, Modular Representation Theory, Number Theory, Real Analysis, Measure Theory Euclidean Space, Numerical Analysis, Category Theory.

PREPRINTS

The Monoid Of Binary Relations On A Set Of Size Four Has Infinite Representation Type

Spring 2026

The problem of determining the representation type of the full transformation monoid was resolved by Ponizovskii, Putcha, and Ringel. In this paper, we present a similar result for the monoid of binary relations, a partial result for the monoid of Λ -generalized correspondences, and a sufficient condition for a monoid to have infinite representation type.

RESEARCH

Large Microvascular System Solutions

Fall 2019

Metropolitan State University of Denver | Mentor: Dr. Brendan Fry

Investigating what factors of a microvascular system control pressures and flows throughout a network. Using tools from linear algebra, analysis, and homology.

Representation Theory and Galois Automorphism

Spring 2020

Metropolitan State University of Denver | Mentor: Dr. Mandi Schaeffer Fry

Investigating the connection between Galois Automorphism and characters of groups.

Cartan Matrix of Boolean Matrix Algebra

Fall 2022 – Spring 2024

University of Arizona | Mentor: Dr. Klaus Lux

Understanding the structure of the representation of the Boolean matrix algebra.

Extracting Kidney Information from Forms

Fall 2022 – Spring 2023

University of Arizona | Mentor: Dr. Marek Rychlik

Help match donated kidneys to patients in need by improving the discovery of viable kidneys through machine vision.

RELEVANT EXPERIENCE

Semigroup Package Contributor

August 2024 – Present

Open-source project | Managed by James Mitchell, University of St Andrews

The semigroup package is an addition to another open-source project called GAP (a system for computational discrete algebra). The semigroup package extends GAP's functionality to semigroups.

Graduate Teaching Assistant

August 2021 – Present

Department of Mathematics, University of Arizona

Supported different instructors with various teaching styles achieve success in teaching math courses consisting of up to 90 students.

Affiliate Faculty

August 2020 – May 2021

Department of Mathematical and Computer Sciences, Metropolitan State University of Denver

Taught 6 algebra lab courses in Fall 2020 and 4 in Spring 2021. Received extensive training for online classes and worked effectively with corequisite class instructors.

Linear Algebra and Calculus 3 Learning Assistant

January 2020 – July 2020

Department of Mathematical and Computer Sciences, Metropolitan State University of Denver

Tutor / Teacher

2018 – Present

Family Friends | Arizona, Utah, and Colorado

Mathematics Tutor

August 2019 – July 2020

Department of Mathematical and Computer Sciences, Metropolitan State University of Denver

PRESENTATIONS

Cartan Matrix Implementation in Semigroups

May 4, 2026

GAP Days Spring 2026, University of Porto, Porto, Portugal

Various Talks Given

Spring 2022 – Fall 2024

Graduate Number Theory Seminar, University of Arizona

Topics: Introduction to Schemes, Hyperfunctions, Category Theory, Separable Algebras and Tensoring, Canonical Heights on Abelian Varieties, Matrix Similarity in $M_{2,2}(\mathbf{Z})$, Polynomial Parameterization of $SL_2(\mathbf{Z})$, Two-part talk: Serre's GAGA Principle, Computing the j-invariant of orders in imaginary 2 quadratic fields with class number 1.

ChatGPT Impact

Spring 2023

RTG / Instructors Colloquium, University of Arizona

Facilitated several talks among teachers about the capabilities of ChatGPT and responses to student use across combined sessions.

Large Microvascular System Solutions (Poster Presentation)

January 17, 2020

Joint Mathematics Meetings, Colorado Convention Center, Denver

Exactness in Chain Complexes and Homology (Poster Presentation)

April 30, 2019

*Undergraduate Poster Competition, Metropolitan State University of Denver***Event Funding Proposal (15-Minute Talk)**

Fall 2019

*Student Travel at MSU Denver, Metropolitan State University of Denver***OUTREACH AND VOLUNTEER WORK**

Volunteer, Red Cross Blood and Platelet Donations

Fall 2022 – Present

There is a great need for Platelet donations. If you have time and are able, please consider donating.

Volunteer, Tucson Festival of Books

March 2023, 2024, 2026

*Math Department Booth, University of Arizona***Volunteer, United States of America Mathematical Olympiad Proctor**

March 4, 2026

Volunteer, Math Day

Spring 2019

*Department of Mathematical and Computer Sciences, Metropolitan State University of Denver***Speaker Presenter, 15th Annual SIAM Front Range Applied Math Student Conference**

March 2019

*University of Colorado Denver***COMPETITIONS AND AWARDS**

William Lowell Putnam Mathematical Competition

Metropolitan State University of Denver

Spring 2018, 2019

Undergraduate Poster Competition

Metropolitan State University of Denver

April 2019

CONFERENCES ATTENDED

Gap Days Spring 2026

University of Porto, Porto, Portugal

May 2026

Winter School

University of Arizona, Tucson, Arizona, USA

March 2022

Joint Math Meetings

American Mathematical Society, Denver, Colorado

January 2020

15th Annual SIAM Front Range Applied Mathematics Student Conference

University of Colorado Denver

March 2019

PROFESSIONAL ASSOCIATIONS & MEMBERSHIPS

American Mathematical Society

Member

Active

Mathematical Association of America

Departmental Student Nominee

Previously Held